

Sharp pointwise lower bounds for Lebesgue functions

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Abstract

For each n , let λ_n be the Lebesgue function for polynomial interpolation at an arbitrary set of n distinct nodes in $[-1, 1]$. We prove that there are a fixed point $x \in (-1, 1)$ and a constant C such that $\lambda_n(x) > (2/\pi) \log n - C$ for infinitely many n . We also prove that $\limsup_{n \rightarrow \infty} \lambda_n(x) / \log n \geq 2/\pi$ for almost every $x \in (-1, 1)$. These conclusions answer both questions in Erdős Problem 1132. The first proof combines Tao's local potential estimates with a local Riesz differentiation formula, an energy estimate for nodal derivative jumps, and a second moment argument. The second proof uses positive Cauchy transforms and harmonic measure.

1 Introduction

For each $n \geq 1$, let

$$-1 \leq x_{1,n} < \cdots < x_{n,n} \leq 1.$$

Define the node polynomial, the cardinal polynomials, and the Lebesgue function by

$$P_n(x) = \prod_{k=1}^n (x - x_{k,n}),$$
$$\ell_{k,n}(x) = \frac{P_n(x)}{(x - x_{k,n})P'_n(x_{k,n})}, \quad \lambda_n(x) = \sum_{k=1}^n |\ell_{k,n}(x)|.$$

The cardinal polynomials are interpreted by continuity at the nodes. The rows of the array need not be nested.

Theorem 1. *For every triangular array of distinct nodes as above, the following statements hold.*

(i) *There exist $x \in (-1, 1)$ and a finite constant C such that*

$$\lambda_n(x) > \frac{2}{\pi} \log n - C \tag{1.1}$$

for infinitely many n .

(ii) *For Lebesgue almost every $x \in (-1, 1)$,*

$$\limsup_{n \rightarrow \infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}. \tag{1.2}$$

Erdős posed both questions for triangular arrays in [2, p. 68], and they are recorded as Problem 1132 in [1]. The constant in (1.1) is independent of n ; it may depend on the array and the point x .

Tao [3, Theorem 1.10(i)] proved that every fixed nondegenerate interval $I \subset [-1, 1]$ satisfies

$$\sup_{x \in I} \lambda_n(x) \geq \frac{2}{\pi} \log n - O_I(1).$$

His Corollary 1.11 and Remark 1.12 give, for every prescribed $\omega(n) \rightarrow \infty$, a dense, indeed comeager, set of points x for which

$$\lambda_n(x) \geq \frac{2}{\pi} \log n - \omega(n)$$

infinitely often. Theorem 1(i) gives a fixed additive constant at a fixed point. Theorem 1(ii) establishes the sharp normalized lower bound on a set of full measure.

For the first assertion, a Baire category reduction gives an interval on which $\lambda_n \leq (2/\pi) \log n + O(1)$, if the assertion is false. On that interval, Tao's potential estimates and a local Riesz formula control derivatives of all interpolants with bounded nodal data. A positive definite kernel then gives a sharp lower average for the derivative jumps of λ_n at the nodes. This produces linearly many separated points with values at least $(2/\pi) \log n - O(1)$. A second moment argument yields a set of positive measure on which this lower bound recurs infinitely often.

The proof of the almost-everywhere assertion is separate. Normalized barycentric weights define two positive Cauchy transforms whose difference is small above the real line. Harmonic measure transfers a hypothetical upper bound on a fixed measurable set to an inequality for a positive convolution operator. Symmetry and the approximate identity property contradict that inequality.

Throughout the paper, all logarithms are natural and $|E|$ denotes Lebesgue measure. We write $c_0 = 2/\pi$, and suppress the row index when discussing a single row. Constants in local estimates may depend on the fixed intervals and test functions, but not on n or on interpolation data bounded by one. The notation $J \Subset U$ means that J is a compact subset of the open set U .

2 A local differentiation formula

We first record a local form of Riesz's differentiation formula, including the consequence of near equality that will be needed below.

Lemma 2. *Suppose that F is holomorphic in a neighborhood of the square $\{|\Re z|, |\Im z| \leq T\}$, is real on its real diameter, and satisfies*

$$|F(u + iv)| \leq e^{|v|}.$$

For sufficiently large T ,

$$|F'(0)| \leq 1 + O(T^{-1}). \tag{2.1}$$

If $F'(0) \geq 1 - \delta$, where $\delta \geq 0$, then

$$F(\pi/2) \geq 1 - \frac{\pi^2}{4} \delta - O(T^{-1}). \tag{2.2}$$

The constants are absolute.

Proof. Choose $R = m\pi$ with $T/2 \leq R < T$, and integrate

$$\frac{F(z)}{z^2 \cos z}$$

around the square of radius R . On its vertical sides, $|\cos z| = \cosh(\Im z)$; on its horizontal sides, $|\cos z| \geq \sinh R$. The contour integral, divided by $2\pi i$, is consequently $O(R^{-1})$. Taking residues at zero and at $t_j = (j + \frac{1}{2})\pi$ gives

$$F'(0) = \sum_{|t_j| < R} \frac{(-1)^j F(t_j)}{t_j^2} + O(T^{-1}). \quad (2.3)$$

Since $|F(t_j)| \leq 1$ and

$$\sum_{j \in \mathbb{Z}} \frac{1}{t_j^2} = 1,$$

equation (2.1) follows. Isolating the term $t_0 = \pi/2$ in (2.3) gives

$$1 - \delta \leq \frac{4}{\pi^2} F(\pi/2) + 1 - \frac{4}{\pi^2} + O(T^{-1}),$$

which proves (2.2). □

We next apply this formula to interpolants with bounded nodal data.

Lemma 3. *Let $I \Subset (-1, 1)$ be a fixed nondegenerate closed interval. Suppose that, for all sufficiently large n ,*

$$\lambda_n(x) \leq \Lambda_n := c_0 \log n + C \quad (x \in I), \quad (2.4)$$

where C is fixed. There are functions ρ_n on $\text{int } I$ whose restrictions to each fixed compact interval are bounded above and below by positive constants and have uniformly bounded Lipschitz norms. They have the following properties on every fixed interval $J \Subset \text{int } I$.

For every real polynomial

$$q(x) = \sum_{k=1}^n a_k \ell_{k,n}(x), \quad |a_k| \leq 1,$$

uniformly for $x \in J$,

$$|q'(x)| \leq \pi n \rho_n(x) (\Lambda_n + o(1)), \quad |q''(x)| \leq C_J n^2 \log n. \quad (2.5)$$

Moreover, for every fixed D , if

$$\frac{|q'(x)|}{\pi n \rho_n(x)} \geq c_0 \log n - D, \quad (2.6)$$

then there is $y = x + O_J(n^{-1})$ such that

$$\lambda_n(y) \geq c_0 \log n - D', \quad (2.7)$$

where D' depends on I, J, C, D , but not on n, x, q . The point y lies in any fixed compact interior interval containing J in its interior, for all sufficiently large n .

Finally, if h_n have support in a fixed compact subset of $\text{int } I$ and have uniformly bounded Lipschitz norms, then

$$\frac{1}{n} \sum_k h_n(x_{k,n}) = \int h_n(x) \rho_n(x) dx + O(n^{-1/10}). \quad (2.8)$$

Proof. We apply Tao [3, Theorem 4.1] to the interval I in (2.4), and use its conclusions on fixed compact subintervals of $\text{int } I$. For the derivative estimates, choose such an interval containing J in its interior; for (2.8), choose one containing the supports of the functions h_n . Put

$$S_n = \sum_k \frac{1}{|P'_n(x_{k,n})|}, \quad \alpha_n = \frac{1}{n} \log S_n, \quad U_n(z) = \frac{1}{n} \log \frac{1}{|P_n(z)|}.$$

That theorem supplies the asserted bounds on ρ_n , together with

$$U_n(x + i\eta) = \alpha_n - \pi\eta\rho_n(x) + O\left(\frac{\log n}{n} + \eta^3\right) \quad (2.9)$$

for $\eta \geq \log n/n$, uniformly on fixed compact interior intervals. It also gives, for every subinterval K of such an interval,

$$\frac{1}{n} \#\{k : x_{k,n} \in K\} = \int_K \rho_n(x) dx + O\left(\frac{|K|}{A} + \frac{A^2 \log n}{n}\right) \quad (A \geq 1). \quad (2.10)$$

Taking $A = n^{1/10}$ and integrating by parts proves (2.8). The bounds are uniform for the stated class of h_n .

For complex z with $\Re z$ in a fixed compact subinterval of $\text{int } I$ and $\Im z = \eta \geq \log n/n$, the interpolation formula and (2.9) imply

$$|q(z)| \leq \frac{S_n |P_n(z)|}{\eta} \leq \frac{1}{\eta} \exp\left(\pi n \eta \rho_n(\Re z) + O(\log n + n\eta^3)\right). \quad (2.11)$$

Fix $x \in J$, and set

$$L = n^{-1/5}, \quad H = n^{-2/5}, \quad \omega = \pi n \rho_n(x)(1 + D_0 n^{-1/10}),$$

where $D_0 > 0$ is fixed and sufficiently large. On the upper edge of the rectangle

$$|\Re z - x| \leq L, \quad 0 \leq \Im z \leq H,$$

the Lipschitz bound for ρ_n and (2.11) give

$$|q(z)| \leq \Lambda_n e^{\omega H}$$

for all sufficiently large n . Indeed, the allowance in ωH has order $n^{1/2}$, whereas the density variation contributes $O(n^{2/5})$, and the remaining terms contribute $O(\log n)$.

On the lower edge, $|q| \leq \lambda_n \leq \Lambda_n$. Expanding q in Chebyshev polynomials on I also gives the bound $|q(z)| \leq \Lambda_n \exp(O_I(n))$ on the vertical sides. Apply the local Phragmén–Lindelöf estimate [3, Lemma 2.5] to

$$\Lambda_n^{-1} q(z) e^{i\omega(z-x)}.$$

For $|u| \leq L/2$ and $0 \leq v \leq H$, it yields

$$|q(x + u + iv)| \leq \Lambda_n (1 + n^{-10}) e^{\omega|v|}. \quad (2.12)$$

Here the error from the vertical sides is $\exp(O(ne^{-cn^{1/5}}))$; reflection supplies the estimate for negative v .

After the change of variable $t = \omega(z - x)$, the function

$$F(t) = \frac{q(x + t/\omega)}{\Lambda_n(1 + n^{-10})}$$

satisfies Lemma 2 on a square of radius $T \asymp_J n^{3/5}$. Equation (2.1), and the fact that $\Lambda_n n^{-1/10} = o(1)$, give the first estimate in (2.5). Cauchy's estimate on a disk of radius comparable to n^{-1} in (2.12) gives the second estimate.

Under (2.6), replacing q by $-q$ if necessary, we have

$$F'(0) \geq 1 - \frac{C_1}{\Lambda_n}$$

with C_1 independent of n, x, q . Equation (2.2) now shows that

$$|q(x + \pi/(2\omega))| \geq \Lambda_n - C_2.$$

Since $|q(y)| \leq \lambda_n(y)$ for real y , this proves (2.7). $\square$

3 Derivative jumps and a Riesz energy

At each internal interpolation node, write

$$J_k = \lambda'_n(x_{k,n}+) - \lambda'_n(x_{k,n}-).$$

For convenience, the same expression may be defined at the first and last nodes by using the exterior polynomial pieces. Set

$$\nu_k = \frac{1}{S_n |P'_n(x_{k,n})|}.$$

The absolute values of all cardinal polynomials other than $\ell_{k,n}$ have a cusp at $x_{k,n}$, whereas $\ell_{k,n}$ is positive near that node. Consequently

$$J_k = 2 \sum_{j \neq k} |\ell'_{j,n}(x_{k,n})| = 2 \sum_{j \neq k} \frac{\nu_j}{\nu_k |x_{k,n} - x_{j,n}|}. \quad (3.1)$$

Lemma 4. *Under the hypotheses of Lemma 3, choose a nonzero nonnegative smooth function f compactly supported in $\text{int } I$. Then*

$$\sum_k \frac{f(x_{k,n})^2}{n \rho_n(x_{k,n})} \frac{J_k}{2\pi n \rho_n(x_{k,n})} \geq c_0 \log n \int f(x)^2 dx - O(1). \quad (3.2)$$

Terms with $f(x_{k,n}) = 0$ are taken to be zero; ρ_n is only needed on a fixed interior neighborhood of the support of f .

Proof. Put $a_k = f(x_{k,n})/\rho_n(x_{k,n})$, with $a_k = 0$ outside the support of f , and define

$$\mu_n = \frac{1}{n} \sum_k a_k \delta_{x_{k,n}}, \quad K_\varepsilon(t) = \frac{1}{\sqrt{t^2 + \varepsilon^2}}.$$

The kernel K_ε is positive definite. For example, its representation

$$K_\varepsilon(t) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-s\varepsilon^2} e^{-st^2} ds$$

expresses it as a positive mixture of Gaussian kernels. Write

$$\mathcal{E}_\varepsilon(\sigma, \tau) = \iint K_\varepsilon(x - y) d\sigma(x) d\tau(y)$$

for its bilinear energy. Positivity applied to $\mu_n - f(x) dx$ gives

$$\mathcal{E}_\varepsilon(\mu_n, \mu_n) \geq 2\mathcal{E}_\varepsilon(\mu_n, f dx) - \mathcal{E}_\varepsilon(f dx, f dx). \quad (3.3)$$

Take $\varepsilon = 1/n$. Subtracting $f(x)$ in the integral over $|t| \leq 1$, and using the Lipschitz bound for f , gives the uniform estimate

$$(K_{1/n} * f)(x) = 2 \log n f(x) + O_f(1). \quad (3.4)$$

The mass of μ_n is bounded. Also, by (2.8),

$$\int f d\mu_n = \frac{1}{n} \sum_k \frac{f(x_{k,n})^2}{\rho_n(x_{k,n})} = \int f^2 + O(n^{-1/10}). \quad (3.5)$$

Equations (3.3)–(3.5) therefore yield

$$\mathcal{E}_{1/n}(\mu_n, \mu_n) \geq 2 \log n \int f^2 - O(1). \quad (3.6)$$

The diagonal part of this energy is $n^{-1} \sum_k a_k^2 = O(1)$. Since $K_{1/n}(t) \leq 1/|t|$ for $t \neq 0$,

$$\frac{1}{n^2} \sum_{k \neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq 2 \log n \int f^2 - O(1). \quad (3.7)$$

Finally,

$$a_k^2 \frac{\nu_j}{\nu_k} + a_j^2 \frac{\nu_k}{\nu_j} \geq 2a_k a_j.$$

By (3.1), the left side of (3.2) is

$$\frac{1}{\pi n^2} \sum_{k \neq j} \frac{a_k^2 \nu_j}{\nu_k |x_{k,n} - x_{j,n}|}.$$

Pairing the terms indexed by (k, j) and (j, k) , and then applying (3.7), bounds this expression below by

$$\frac{1}{\pi n^2} \sum_{k \neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq \frac{2}{\pi} \log n \int f^2 - O(1).$$

This proves (3.2). □

4 A fixed point with bounded additive loss

Proposition 5. *Assume the local upper bound (2.4) on a fixed nondegenerate interval $I \Subset (-1, 1)$. There are a constant D and a measurable set $G \subset \text{int } I$, with $|G| > 0$, such that every $x \in G$ satisfies*

$$\lambda_n(x) \geq c_0 \log n - D$$

for infinitely many n .

Proof. Fix nested compact interior intervals and a function f as in Lemma 4. Let

$$w_k = \frac{f(x_{k,n})^2}{n\rho_n(x_{k,n})}, \quad T_k = \frac{J_k}{2\pi n\rho_n(x_{k,n})}, \quad W = \int f^2 > 0.$$

By (2.8),

$$\sum_k w_k = W + O(n^{-1/10}), \quad 0 \leq w_k \leq C_f/n. \quad (4.1)$$

On each interval between successive nodes, λ_n equals an interpolant with nodal data in $\{-1, 1\}$. Both one-sided derivatives at a node therefore satisfy (2.5), so

$$0 \leq T_k \leq \Lambda_n + o(1) \quad (4.2)$$

uniformly on the support of f . Lemma 4, (4.1), and (4.2) show that

$$\sum_k w_k(\Lambda_n + 1 - T_k) = O(1), \quad (4.3)$$

with nonnegative summands for all sufficiently large n . It follows that there is a fixed D for which the nodes in the support of f satisfying

$$T_k \geq c_0 \log n - D \quad (4.4)$$

carry at least $W/2$ of the weights in (4.1). In particular, there are at least κn such nodes, with $\kappa > 0$ independent of n . Discarding the first and last nodes does not change this conclusion after reducing κ .

At a node satisfying (4.4), at least one adjacent polynomial piece q has

$$|q'(x_{k,n})| \geq \pi n \rho_n(x_{k,n})(c_0 \log n - D).$$

Lemma 3 gives a point

$$y_k = x_{k,n} + O(n^{-1}), \quad \lambda_n(y_k) \geq c_0 \log n - D_1. \quad (4.5)$$

These points cannot all concentrate in a small number of node gaps. To see this, use the adjacent piece with the large derivative. Its values at the two endpoints of that gap are both 1. Rolle's theorem and the second derivative bound in (2.5) imply that the gap has length at least d/n , for a fixed $d > 0$. If the gap leaves the fixed interior neighborhood, its length is already bounded below by a positive constant, so the same conclusion holds.

Any interval of length less than d/n contains at most two nodes having an adjacent gap of length at least d/n . Otherwise a middle node would have both neighboring gaps shorter than d/n . Thus every interval of length $O(n^{-1})$ contains only a bounded number of the nodes in (4.4). The displacement bound in (4.5) gives the same bound, with multiplicities, for the points y_k . A greedy selection supplies at least $\kappa_1 n$ points satisfying (4.5) and separated by d_1/n , for fixed positive κ_1, d_1 .

Equation (2.5), applied to every polynomial piece, also shows that λ_n is $C_I n \log n$ -Lipschitz on a fixed compact interior interval containing these points. Choose a fixed sufficiently small $b > 0$, and let A_n be the union of the intervals of radius

$$r_n = \frac{b}{n \log n}$$

centered at the selected points. For all sufficiently large n , these intervals are disjoint, lie in a fixed compact interior interval, and satisfy

$$\lambda_n(x) \geq c_0 \log n - D_2 \quad (x \in A_n), \quad \frac{c}{\log n} \leq |A_n| \leq \frac{C_2}{\log n}. \quad (4.6)$$

The separation of the centers in the m -th row implies, for every interval B ,

$$|A_m \cap B| \leq C_3 \left(\frac{|B|}{\log m} + \frac{1}{m \log m} \right).$$

Since A_n has at most n components, for $m > n$,

$$|A_n \cap A_m| \leq C_3 \left(\frac{|A_n|}{\log m} + \frac{n}{m \log m} \right). \quad (4.7)$$

Put $B_j = A_{2^j}$. Equations (4.6) and (4.7) give

$$|B_j| \asymp j^{-1}, \quad |B_j \cap B_k| \leq C_4 \left(\frac{1}{jk} + \frac{2^{j-k}}{k} \right) \quad (j < k). \quad (4.8)$$

For every fixed j_0 , the sum of the measures of B_j , $j_0 \leq j \leq N$, grows at least as a positive constant times $\log N$. By (4.8), the integral of the square of the sum of their indicators is $O((\log N)^2)$. Cauchy–Schwarz consequently gives a positive lower bound, independent of j_0 , for

$$\left| \bigcup_{j \geq j_0} B_j \right|.$$

Continuity of measure along these decreasing tail unions shows that

$$|\limsup_{j \rightarrow \infty} B_j| > 0. \quad (4.9)$$

Together with (4.6), this proves the proposition. $\square$

Proof of Theorem 1(i). Suppose the assertion were false. Then

$$\lambda_n(x) - c_0 \log n \longrightarrow -\infty \quad \text{for every } x \in (-1, 1). \quad (4.10)$$

Fix a nondegenerate compact interval $K_0 \subset (-1, 1)$. The closed sets

$$F_M = \{x \in K_0 : \lambda_n(x) \leq c_0 \log n + M \text{ for every } n \geq 2\}, \quad M = 1, 2, \dots,$$

cover K_0 , by (4.10). Baire's theorem supplies a nondegenerate closed subinterval $I \subset K_0$ on which (2.4) holds with a fixed C . Proposition 5 contradicts (4.10). Increasing the final additive constant by one gives the strict inequality in the statement. $\square$

5 Positive Cauchy transforms

We now prove the almost-everywhere assertion. Again suppress the row index, and put

$$S = \sum_k \frac{1}{|P'(x_k)|}, \quad \nu_k = \frac{1}{S|P'(x_k)|}, \quad \epsilon_k = \operatorname{sgn} P'(x_k).$$

Define

$$m(z) = \sum_k \frac{\nu_k}{z - x_k}, \quad g(z) = \sum_k \frac{\epsilon_k \nu_k}{z - x_k} = \frac{1}{SP(z)}. \quad (5.1)$$

The last identity is the partial fraction formula for $1/P$. Its expansion at infinity also gives

$$\sum_k \epsilon_k \nu_k x_k^j = 0 \quad (0 \leq j \leq n-2). \quad (5.2)$$

For $n \geq 2$, both $m + g$ and $m - g$ are Cauchy transforms of positive probability measures: their masses at x_k are $(1 \pm \epsilon_k) \nu_k$.

For real x away from the nodes, write

$$H(x) = \sum_k \frac{\nu_k}{|x - x_k|}, \quad \gamma_s(x) = -\Im m(x + is), \quad p_s(t) = \frac{s}{\pi(t^2 + s^2)}.$$

Then

$$\lambda_n(x) = \frac{H(x)}{|g(x)|}, \quad H(x) = \frac{2}{\pi} \int_0^\infty \gamma_s(x) \frac{ds}{s}, \quad (5.3)$$

and the Poisson semigroup gives

$$\gamma_s = p_{s-h} * \gamma_h \quad (s > h). \quad (5.4)$$

Lemma 6. *If $0 < a < 1$ and $h = n^{-a}$, then*

$$\sup_{x \in [-1, 1]} \frac{|g(x + ih)|}{\gamma_h(x)} \rightarrow 0. \quad (5.5)$$

Proof. The function $t \mapsto (x + ih - t)^{-1}$ is analytic on the Bernstein ellipse of parameter $R = 1 + h/4$, where its modulus is $O(h^{-1})$. Truncating its Chebyshev expansion at degree $n-2$ gives a polynomial p satisfying

$$\sup_{t \in [-1, 1]} \left| \frac{1}{x + ih - t} - p(t) \right| \leq Ch^{-2} e^{-cnh}. \quad (5.6)$$

Indeed, the Chebyshev coefficients are $O(h^{-1}R^{-j})$, and their tail is a geometric series. The signed measure in (5.1) has total variation one and annihilates p , by (5.2). Thus (5.6) bounds $|g(x + ih)|$. On the other hand,

$$\gamma_h(x) = \sum_k \frac{\nu_k h}{(x - x_k)^2 + h^2} \geq \frac{h}{4 + h^2} \quad (x \in [-1, 1]).$$

This proves (5.5). □

Lemma 7. *Let F be the Cauchy transform of a positive probability measure with finite support. For every bounded real interval S_0 , every $x \in \mathbb{R}$, and every $h > 0$,*

$$\int_{\mathbb{R}} p_h(x - y) \mathbf{1}_{\{F(y) \in S_0\}} dy = \frac{1}{\pi} \int_{S_0} \frac{-\Im F(x + ih)}{(t - \Re F(x + ih))^2 + (\Im F(x + ih))^2} dt. \quad (5.7)$$

Proof. Compose harmonic measure of S_0 in the lower half-plane with F . The resulting bounded harmonic function on the upper half-plane has the indicated boundary values almost everywhere. The exceptional boundary points are poles of F and preimages of the endpoints of S_0 , a finite set. Its Poisson representation is (5.7). □

6 The almost-everywhere bound

Proof of Theorem 1(ii). Suppose that the assertion fails. By a countable-union argument, there are a measurable set $E \subset (-1, 1)$ of positive measure, a constant $0 < c < c_0$, and an integer N such that

$$\lambda_n(y) \leq c \log n \quad (y \in E, n \geq N). \quad (6.1)$$

Choose constants

$$K > 1, \quad 0 < b < a < 1, \quad \frac{2b}{\pi c K} > 1,$$

which is possible because $c < 2/\pi$. Set

$$h = n^{-a}, \quad \eta = n^{-b}, \quad H^\eta(x) = \frac{2}{\pi} \int_\eta^1 \gamma_s(x) \frac{ds}{s}.$$

By (5.3), $H^\eta \leq H$. Direct differentiation gives $|\gamma'_s| \leq \gamma_s/s$, and consequently

$$|(\log H^\eta)'| \leq \eta^{-1}. \quad (6.2)$$

Fix $x \in E$, and write

$$u = \Re m(x + ih), \quad \gamma = \gamma_h(x), \quad S_x = [u - K\gamma, u + K\gamma].$$

By (5.5) and (5.7), each of the two events

$$m(y) + g(y) \in S_x, \quad m(y) - g(y) \in S_x$$

has probability

$$p_K + o(1), \quad p_K = \frac{2}{\pi} \arctan K > \frac{1}{2},$$

under the Poisson probability $p_h(x - y) dy$. The error is uniform in $x \in E$. Their intersection therefore has probability at least

$$q_K + o(1), \quad q_K = 2p_K - 1 > 0. \quad (6.3)$$

Choose R so that the Poisson mass of $|y - x| > Rh$ is less than $q_K/8$, and let

$$E_n = \{x \in E : (p_h * \mathbf{1}_{\mathbb{R} \setminus E})(x) < q_K/8\}.$$

The approximate identity theorem gives $|E \setminus E_n| \rightarrow 0$. For all sufficiently large n , (6.3) implies that, for every $x \in E_n$, there exists $y \in E$ with $|y - x| \leq Rh$ and both events true. We may take y away from the nodes. At this point,

$$|g(y)| \leq K\gamma_h(x).$$

Using (5.3), (6.1), and (6.2), we obtain

$$H^\eta(x) \leq e^{Rh/\eta} H^\eta(y) \leq cK e^{Rh/\eta} (\log n) \gamma_h(x) \quad (x \in E_n). \quad (6.4)$$

Let T_n be convolution with the symmetric probability kernel

$$k_n(t) = \frac{1}{b \log n} \int_\eta^1 p_{s-h}(t) \frac{ds}{s}.$$

Equation (5.4) shows that

$$H^\eta = \frac{2b}{\pi}(\log n)T_n\gamma_h.$$

Thus (6.4) becomes

$$\frac{T_n\gamma_h(x)}{\gamma_h(x)} \leq \frac{1}{A_n} \quad (x \in E_n), \quad A_n = \frac{2b}{\pi cK} e^{-Rh/\eta} \longrightarrow A > 1. \quad (6.5)$$

Integrate (6.5) over E_n , discard the contribution from $y \notin E_n$, and symmetrize. Positivity of γ_h , symmetry of k_n , and $(r + r^{-1})/2 \geq 1$ give

$$\frac{|E_n|}{A_n} \geq \iint_{E_n \times E_n} k_n(x-y) \frac{\gamma_h(y)}{\gamma_h(x)} dy dx \geq \iint_{E_n \times E_n} k_n(x-y) dy dx. \quad (6.6)$$

The kernels k_n form an approximate identity. In fact, for every fixed $\delta > 0$,

$$\int_{|t|>\delta} k_n(t) dt \leq \frac{C}{b \log n} \int_\eta^1 \min(1, s/\delta) \frac{ds}{s} \longrightarrow 0.$$

It follows that

$$\iint_{E \times E} k_n(x-y) dy dx \longrightarrow |E|.$$

Replacing E by E_n changes this integral by at most $2|E \setminus E_n|$. Taking limits in (6.6) therefore yields

$$\frac{|E|}{A} \geq |E|,$$

contrary to $|E| > 0$ and $A > 1$. This proves the assertion. $\square$

AI tool disclosure

The proofs and manuscript were developed using GPT-6 Astra (OpenAI). The author is solely responsible for the mathematical content of the paper.

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